

Course Outline

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I. APPROACH

0.1 System Analysis

1. Fundamental Systems Concepts and Perspectives

(a) System Components and Characterizations

(b) Systems Science

i. classical (calculus-based) open/closed systems

ii. set theory

iii. graph theory

iv. network theory

v. cybernetics

vi. information theory

vii. game theory

viii. complexity theory

ix. decision theory

x. queuing theory

xi. theory of structure (the order of parts) and function (the order of processes)

(c) Mental Models

2. Spatial Scales

(a) Global

(b) Regional

(c) Local

3. Temporal Scales

(a) Past

(b) Present

(c) Future

4. Analytics

(a) Probability Models

- (b) Statistical Models
- (c) Agent-Based Models
- (d) Time Series Models

0.2 Opportunity

1. External

- (a) Creating New Systems
- (b) Re-vamping Existing Systems
- (c) The Critical Systems
 - i. Economic
 - ii. Governance
 - iii. Cultural
 - iv. Artistic
 - v. Environmental
 - vi. Existential
 - A. Food
 - B. Water
 - C. Clothing
 - D. Shelter

2. Internal

- (a) Creativity
- (b) Intuition
- (c) Mind, Body, Spirit

II. TOPICS

0.3 Sustainability Concepts and Frameworks

1. Sustainability Concepts

- (a) Brundtland Commission
- (b) UN Sustainable Development Goals
- (c) Marshall and Toffel

2. Sustainability Frameworks

- (a) Ecological Sustainability

- i. Ecosystem Services: Definitions and Metrics
 - ii. Global Ecosystem Health and Ecosystem Service Delivery
- (b) Social Sustainability
- (c) Economic Sustainability
- (d) Agricultural Sustainability

0.4 A Sustainable Biosphere

1. Functional Terrestrial and Aquatic Ecosystems

- (a) Necessary Inputs
 - i. Physical Resources
 - A. Space
 - B. Time
 - C. Flow
 - ii. Biological Resources
 - A. Reliable Primary Productivity
- (b) Necessary Outputs
 - i. Measurable Properties
 - A. Biodiversity
 - B. Energy and Material Recycling
 - ii. Emergent Properties
 - A. Adaptive Capacity
 - B. Resilience

0.5 A Sustainable Anthroposphere

1. Culture

- (a) Pillars of Belief
 - i. Reality and Imagination
 - ii. Identity Legacy
- (b) Open Cooperative Competition
 - i. Cohesion
 - ii. Energy
 - iii.

2. Power

- (a) Governance

(b) Elites

3. Money

(a) Value vs Money vs Wealth

(b) Economy and Trade

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